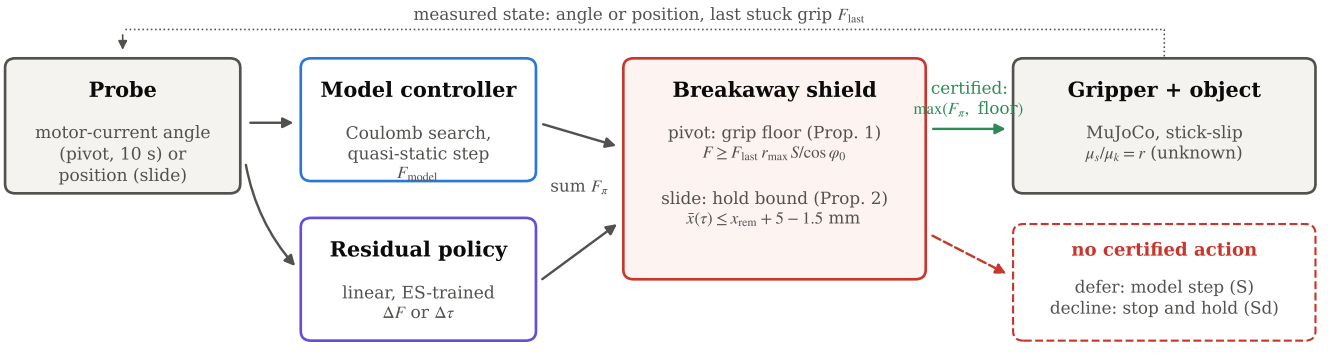
measured state: angle or position, last stuck grip F_{last}
Probe
motor-current angle (pivot, 10 s) or position (slide)
Model controller
Coulomb search, quasi-static step
F_{model}
Residual policy
linear, ES-trained
ΔF or $\Delta\tau$
sum F_π
Breakaway shield
pivot: grip floor (Prop. 1)
$F \geq F_{\text{last}}\, r_{\max}\, S/\cos\varphi_0$
slide: hold bound (Prop. 2)
$\bar{x}(\tau) \leq x_{\text{rem}} + 5 - 1.5$ mm
certified:
$\max(F_\pi,\ \text{floor})$
Gripper + object
MuJoCo, stick-slip
$\mu_s/\mu_k = r$ (unknown)
no certified action
defer: model step (S)
decline: stop and hold (Sd)